THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 1287	$99 \times 13 = (100 - 1) \times 13$
2	**(a)** 2 : 3	Divide both terms by 6
3	**(a)** 25%	Multiply by 100: $\frac{1}{4} \times 100 = 25$
4	**(c)** 5	$3^2 + 4^2 = 9 + 16 = 25$, hypotenuse $= 5$
5	**(d)** 2:3	Divide both terms by HCF 6
6	**(c)** 91	$7 \times 10 + 7 \times 3$
7	**(c)** 12	Multiply both terms by 4
8	**(d)** $\frac{1}{5}$	$20\% = \frac{20}{100} = \frac{1}{5}$
9	**(a)** 4	$5^2 - 3^2 = 16$, other leg $= 4$
10	**(a)** 12	Multiplier 4, so $x = 3 \times 4$
11	**(b)** $3x + 12$	Multiply each term inside by the outside number
12	**(a)** Yes	2x12 = 24 and 3x8 = 24
13	**(b)** 25	10/100 × 250
14	**(c)** 3, 4, 5	$3^2 + 4^2 = 25 = 5^2$
15	**(a)** 250	One part $= 400 \div 8 = 50$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$99 \times 16 = (100 - 1) \times 16$ (1) $= 1600 - 16 = 1584$ (1).	1+1
17	2 m = 200 cm (1); ratio $= 1 : 5$ (1).	1+1
18	$\frac{3}{8} \times 100 = 37.5\%$ (2).	2
19	$h^2 = 3^2 + 4^2 = 9 + 16 = 25$ (1); $h = 5$ cm (1).	1+1
20	Cost of 1 pen $=$ ₹15 (1); cost of 15 pens $=$ ₹225 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$2x + 6 + 4x + 20$ (1); collect like terms $= 6x + 26$ (2).	1+2
22	Total parts 8 (1); one part ₹ 150 (1); shares ₹ 450 and ₹ 750 (1).	1+1+1
23	Profit = ₹ 50 (1); percent = 50/200×100 (1) = 25% (1).	1+1+1
24	Height$^2 = 5^2 - 3^2 = 25 - 9 = 16$ (2); height $= 4$ m (1).	2+1
	OR: Actual distance $= 5 \times 12 = 60$ km (2); map distance $= 96 \div 12 = 8$ cm (1).	2+1
25	$x^2 + 3x + 2x + 6$ (2); $= x^2 + 5x + 6$ (1).	2+1
	OR: Product of extremes 3x20$= 60$ (1); product of means 4x15$= 60$ (1); equal, so in proportion (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$x^2 + 4x + 2x + 8$ (2); $= x^2 + 6x + 8$ (1); at $x = 3$: $9 + 18 + 8 = 35$ (2).	2+1+2
	OR: Total parts 8, one part $= 6$ (2); numbers 18 and 30 (2); difference 12 (1).	2+2+1
27	Food = 30% of 20000 = ₹ 6000 (2); rent = 25% of 20000 = ₹ 5000 (2); left = 20000 − 11000 = ₹ 9000 (1).	2+2+1
	OR: Diagonal$^2 = 9 + 16 = 25$, diagonal $= 5$ m (3); distance along two sides $= 7$ m, saved $= 2$ m (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $a^2 + 2ab + b^2$	Standard identity
2	**(c)** 1 : 4	Convert to the same unit first
3	**(a)** 550	Increase = 50; new = 550
4	**(c)** 12	Hypotenuse $= 5$; perimeter $= 3 + 4 + 5 = 12$
5	**(d)** 35	5×7
6	**(a)** 441	$21^2 = 21 \times 21$
7	**(c)** ₹ 250	One part $= 400 \div 8 = 50$
8	**(d)** 100	SI = PRT/100
9	**(a)** 5	Diagonal$^2 = 9 + 16 = 25$
10	**(a)** 36	10% of $360^\circ = 36^\circ$
11	**(c)** 17	$3 \times 4 + 5$
12	**(b)** ₹ 8	$48 \div 6$
13	**(b)** 900	SP = MP − discount
14	**(a)** hypotenuse	Side opposite the right angle
15	**(d)** 6	Workers × days is constant: $4 \times 12 = 48$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$7 \times x + 7 \times 3$ (1) $= 7x + 21$ (1).	1+1
17	Both terms of the first ratio are multiplied by 3 (1); $x = 27$ (1).	1+1
18	$12/100 \times 500 =$ ₹ 60 (2).	2
19	$x^2 = 17^2 - 8^2 = 289 - 64 = 225$ (1); $x = 15$ cm (1).	1+1
20	Men × days $= 10 \times 24 = 240$ (1); days $= 240 \div 8 = 30$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Discount = 10/100×1200 = ₹ 120 (2); SP = 1200 − 120 = ₹ 1080 (1).	2+1
22	Hypotenuse$^2 = 25 + 144 = 169$, hypotenuse $= 13$ cm (2); perimeter $= 5 + 12 + 13 = 30$ cm (1).	2+1
23	Total parts $= 10$, one part $=$ ₹120 (1); A = ₹ 240 (1); B = ₹ 360, C = ₹ 600 (1).	1+1+1
24	$24 = 20 + 4$ (1); $20^2 + 2(20)(4) + 4^2 = 400 + 160 + 16$ (1); $= 576$ (1).	1+1+1
	OR: Cost per kg $= 225 \div 5 = 45$ (1); cost of 8 kg $= 45 \times 8$ (1); $=$ ₹360 (1).	1+1+1
25	SI = 4000×5×2/100 = ₹ 400 (2); amount = P + SI = ₹ 4400 (1).	2+1
	OR: Distance$^2 = 6^2 + 8^2 = 100$ (2); distance $= 10$ km (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Angle per student $= 360^\circ \div 30 = 12^\circ$ (1); Cricket: 144°, Football: 108° (2); Tennis: 72°, Hockey: 36° (1); sum $= 360^\circ$ (1).	1+2+1+1
	OR: $104 \times 25 = (100 + 4) \times 25 = 2500 + 100$ (2); $= 2600$ (1); direct multiplication gives 2600 (2).	2+1+2
27	Cost of 1 pen ₹ 15 (1); cost of 10 pens ₹ 150 (2); pens for ₹ 210: 14 (2).	1+2+2
	OR: (i) SP = 2000×110/100 = ₹ 2200 (2); (ii) SP = 2000×95/100 = ₹ 1900 (2); difference = ₹ 300 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 7	The number multiplying x
2	**(c)** ₹ 210	Find the cost of 1 book first
3	**(a)** 60%	Multiply by 100: $\frac{3}{5} \times 100 = 60$
4	**(d)** 10	$6^2 + 8^2 = 36 + 64 = 100$, hypotenuse $= 10$
5	**(d)** 3:5	Divide both terms by HCF 5
6	**(c)** 2424	$101 \times 24 = (100 + 1) \times 24$
7	**(c)** 4 : 7	Divide both terms by 5
8	**(c)** $\frac{7}{20}$	$35\% = \frac{35}{100} = \frac{7}{20}$
9	**(b)** 8	$10^2 - 6^2 = 64$, other leg $= 8$
10	**(b)** 30	Multiplier 6, so $x = 5 \times 6$
11	**(a)** 324	$6 \times 50 + 6 \times 4$
12	**(a)** 30	Multiply both terms by 6
13	**(d)** 90	20/100 × 450
14	**(b)** 6, 8, 10	$6^2 + 8^2 = 100 = 10^2$
15	**(a)** 200	One part $= 360 \div 9 = 40$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$34 = 30 + 4$ (1); $30^2 + 2 \times 30 \times 4 + 4^2 = 1156$ (1).	1+1
17	Cost of 1 notebook $= 91 \div 7 = 13$ (1); cost of 12 $=$ ₹156 (1).	1+1
18	Profit = SP − CP (or CP − SP) (1) = ₹ 60 (1).	1+1
19	$6^2 + 8^2 = 100 = 10^2$ (1), so it is right-angled (1).	1+1
20	Total parts $= 7$, one part $=$ ₹120 (1); A gets ₹ 360 and B gets ₹ 480 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Angle $= 30\% \times 360^\circ = 108^\circ$ (2); students $= 60$ (1).	2+1
22	$3x + 6 + 5x + 5$ (1); collect like terms $= 8x + 11$ (2).	1+2
23	Total parts 7 (1); one part ₹ 120 (1); shares ₹ 240 and ₹ 600 (1).	1+1+1
24	Loss = ₹ 50 (1); percent = 50/500×100 (1) = 10% (1).	1+1+1
	OR: Height$^2 = 10^2 - 6^2 = 100 - 36 = 64$ (2); height $= 8$ m (1).	2+1
25	Actual distance $= 4 \times 30 = 120$ km (2); map distance $= 150 \div 30 = 5$ cm (1).	2+1
	OR: $x^2 + 5x + 4x + 20$ (2); $= x^2 + 9x + 20$ (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Triplet: 4, 3, 5 (2); $16 + 9 = 25$ (2); $5^2 = 25$ so verified (1).	2+2+1
	OR: Total food $= 20 \times 15 = 300$ person-days (2); people now $= 30$ (1); days $= 300 \div 30 = 10$ (2).	2+1+2
27	$x^2 + 5x + 3x + 15$ (2); $= x^2 + 8x + 15$ (1); at $x = 2$: $4 + 16 + 15 = 35$ (2).	2+1+2
	OR: Total parts 9, one part $= 8$ (2); numbers 16 and 56 (2); difference 40 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.